

inherent in an asset that can be taken advantage of by an adversary. On the other hand, a threat arises when an adversary possesses the necessary motivation, resources, and intent to exploit a vulnerability. Finally, the resulting risk signifies the potential for the asset to suffer loss or damage when an adversary actively exploits the identified vulnerability.

Decoding the complexities of cyber risks in vast satellite constellations demands careful consideration of these terms. By recognising the distinct nature of vulnerabilities, threats, and risks, we need to gain a deeper understanding of the intricate dynamics involved. Only then can we devise effective strategies to mitigate potential dangers and safeguard these valuable assets.

In order to ensure the continuity and resilience of an organisation, it is crucial to establish an adaptive system. McCormick defines an adaptive system as one that “aggressively adapts to the unexpected,” emphasising the need for continuous learning from the environment. This principle applies not only to cyber and space systems but also to any critical system, necessitating their autonomy and isolation from the public network to prevent intrusions. This can be achieved through the development of independent information and communications technology (ICT) systems that operate without internet connectivity. Additionally, accrediting all public and classified systems serves as a valuable practice for identifying and protecting potential weak links against intrusions and attacks.

Large satellite constellations represent complex systems spanning multiple domains, sectors, and assets. Technological advancements enabling these constella-



*Satellite in low Earth orbit illustrative*

tions require a coordinated effort to address the challenges associated with spacecraft operations and space traffic. As the significance of these constellations grows, so does the attention from threat actors targeting space systems through cyber-attacks. Consequently, research and intelligence on space system vulnerabilities are becoming more sophisticated, detailed, and accessible. To address these evolving threats, states are working to increase awareness of vulnerabilities and adversary capabilities. The National Air and Space Intelligence Centre (NASIC) has identified cyber threats and risks in four categories: the space segment, user segment, link segment, and ground segment. The supply chain also poses a significant potential point of access for cyber-attacks, given the involvement of multiple manufacturers and system integrators. Military satellites already employ advanced encryption methods and secure ground infrastructures to mitigate the risk of cyber-attacks. Effective risk monitoring begins with a well-designed cyber resilient posture,

where space systems continuously monitor, anticipate, and adapt to mitigate cyber activities that could manipulate, deny, degrade, disrupt, destroy, surveil, or eavesdrop on space operations.

Radiation exposure in Low Earth Orbit (LEO) can lead to electronic component damage and solar panel degradation aboard spacecraft. Collaborative efforts between the European Space Agency (ESA) and NASA aim to establish a pre-warning system based on in-space and Earth monitoring of solar activity, aiming to minimise the impact of adverse space weather on both spacecraft and Earth.

In-space operations and end-of-life considerations also entail risks. As of January 1, 2021, there are 6,542 satellites recorded by the Union of Concerned Scientists (UCS), with 3,372 satellites active and 3,170 satellites inactive. This number is expected to exceed 100,000 by the end of the decade. Despite employing collision avoidance manoeuvres, the implementation of artificial intelligence (AI) can play a vital role in early collision risk detection and